



**SYDNEY BOYS HIGH  
SCHOOL**  
MOORE PARK, SURRY HILLS

**2006**

YEAR 11  
HSC Task 1  
Term 4

# Mathematics

## General Instructions

- Working time – 90 Minutes
- Reading time – 5 Minutes
- Write using black or blue pen
- Board approved calculators may be used.
- All necessary working should be shown in every question if full marks are to be awarded.
- Marks may not be awarded for messy or badly arranged work.
- Hand in your answer booklets in 5 sections. Section A (Question 1), Section B (Question 2), Section C (Question 3), Section D (Question 4) and Section E (Question 5).

## Total Marks – 76

- Attempt Questions 1 – 5.
- All questions are NOT of equal value.

Examiner: *A. Ward*

**SECTION A****Question 1 (16 Marks)****Marks**

**a)** Differentiate with respect to  $x$  and simplify:

(i)  $y = x^5 - 1$  1

(ii)  $y = (3x^4 - 5)^7$  1

(iii)  $y = \frac{x+1}{3-x}$  2

**b)** Write down the third and fourth terms of the series  $12 + 6 + \dots$  if it is:

(i) an arithmetic series 1

(ii) a geometric series 1

**c)** (i) Find, to 2 decimal places, the roots of:

$$2x^2 - 3x - 4 = 0$$
 2

(ii) Show that  $2x^2 - 3x + 4 = 0$  has no real roots. 1

**d)** Determine each of the following:

(i)  $\int x^6 dx$  1

(ii)  $\int (x-1)(x+2)dx$  2

(iii)  $\int \frac{\sqrt{t}+1}{\sqrt{t}} dt$  2

**e)** Three terms of an arithmetic series have the sum 21 and a product of 315.

Find the 3 numbers. 2

**End of Section A**

**SECTION B – Start a new booklet****Marks****Question 2. (17 marks)**

- a)** Given that  $ab^c = d$  :
- (i) Find  $b$  in terms of  $a$ ,  $c$  and  $d$ . 1
  - (ii) Find  $c$  in terms of  $a$ ,  $b$  and  $d$  2
  - (iii) Calculate  $b$ , correct to 4 significant figures, when  
 $a = 75.12$ ,  $c = 1.142$  and  $d = 61.94$  . 1
- b)** Given that,  $f(x) = a - 2x - x^2$  where  $a$  is a constant. Find:
- (i) the value for  $a$  for which the roots of the equation differ by 3. 2
  - (ii) the set of values of  $a$  for which  $f(x) < 0$  for all values of  $x$ . 2
- c)** At what points does the tangent to  $f(x) = 2x^3 - 3x^2 + 1$ , have slope 0. 2
- d)** Evaluate:  $\int_{-1}^2 (3x^2 - 2x) dx$  2
- e)** For the function,  $f(x) = 5x^3 - 7x^2 + 3x + 2$
- (i) Show that  $f(x)$  passes through the point (1,3) 1
- At this point, find:
- (ii) the gradient. 1
  - (iii) the equation of the tangent in gradient-intercept form. 1
  - (iv) the equation of the normal in general form. 2

**End of Section B**

**SECTION C – Start a new booklet****Marks****Question 3. (15 marks)**

- a) Find the values of  $y$  which satisfy the equation:

$$(8^y)^y \times \frac{1}{32^y} = 4 \quad 2$$

- b) A point  $P$  is equidistant from the  $x$ -axis and the point  $F(0,2)$ . Find the locus of the point  $P$ . 3

- c) Using first principles, find the derivative of the function  $f(x) = x^2 + x$ , (all working must be shown). 2

- d) Find the area bounded by  $y = \sqrt{x} + 3$  and the  $x$ -axis for  $1 \leq x \leq 4$ . 2

- e) The equation  $3x^2 - 6x + 8 = 0$  has roots  $\alpha$  and  $\beta$ . Find an equation which has roots  $\frac{1}{\alpha}$  and  $\frac{1}{\beta}$  2

- f) A wall vase has one plane face, and its volume is equivalent to that generated when the area enclosed by  $x = \frac{y^3}{64} + 1$ , the  $y$ -axis and  $y = 8$ , is rotated through two right angles about the  $y$ -axis, the units being centimetres. Calculate its volume. 4

**End of Section C**

**SECTION D – Start a new booklet****Marks****Question 4. (12 marks)**

- a)**
- (i) Tabulate, to 2 decimal places, the values of the function  $f(x) = \sqrt{1+x^2}$ , at **unit** intervals from  $x = 2$  to  $x = 5$  inclusive. 2
- (ii) Use these values to find an estimate, by the trapezoidal rule, of the area between  $y = f(x)$  and the  $x$ -axis for  $2 \leq x \leq 5$  to 3 decimal places.. 2
- b)** A point  $P$  has  $x$ -coordinate  $a$  which is taken to be on the line  $y = 3x - 9$ .
- (i) If  $Q$  is the point  $(1,4)$ , show that  $PQ^2 = 10a^2 - 80a + 170$  2
- (ii) Find the value of  $a$  which will make  $PQ$  a minimum. 2
- (iii)  $N$  is a point on the line such that  $QN$  is perpendicular to the line. Find the co-ordinates of  $N$ . 2
- (iv) Find the equation of  $QN$  in general form. 2

**End of Section D**

**SECTION E – Start a new booklet****Marks****Question 5. (16 marks)**

- a)** Solve for  $x$ :
- (i)  $\log_5 x + \log_2 8 = 0$  1
- (ii)  $\log_3 x + 3\log_x 3 = 4$  2
- b)** Prove that, if the sum of the radii of two circles remains constant, the sum of the areas of the circles is least when the circles are equal. 3
- c)** A prize fund is set up with an investment of \$2000, to provide a prize of \$150 each year. The fund accrues compound interest at 5% p.a. paid six monthly. The first prize is awarded 1 year after the initial investment, after interest is received.
- (i) Find the value of the fund immediately after the first years prize is drawn from the fund. 1
- (ii) Find the value of the fund immediately after the third prize is drawn from the fund. 2
- (iii) Find the number of prizes of the full \$150 which can be drawn from the fund. 3
- d)** Three real, distinct and non-zero numbers  $a$ ,  $b$  and  $c$  are such that  $a$ ,  $b$ ,  $c$  are in arithmetic series and  $a$ ,  $c$ ,  $b$  are in geometric series.
- (i) Find the numerical value of the common ratio of the geometric series. 2
- (ii) Hence, find an expression in terms of  $a$  for the sum to infinity of the geometric series whose first terms are  $a$ ,  $c$ ,  $b$ . 2

**End of Section E****End of Examination Paper**

### Question 1

a) i)  $y = x^5 - 1$   
 $y' = 5x^4$

ii)  $y = (3x^4 - 5)^7$   
 $y' = 7(3x^4 - 5)^6 \cdot 12x^3$   
 $= 84x^3(3x^4 - 5)^6$

iii)  $y = \frac{x+1}{3-x}$        $u = x+1$     $v = 3-x$   
 $u' = 1$     $v' = -1$        $y' = \frac{vu' - uv'}{v^2}$

$$y' = \frac{(3-x) + (x+1)}{(3-x)^2}$$
$$= \frac{4}{(3-x)^2}$$

b) i)  $a = 12, d = -6$

$$\therefore T_3 = 0, T_4 = -6$$

ii)  $a = 12, r = \frac{1}{2}$

$$\therefore T_3 = 3, T_4 = \frac{3}{2}$$

c) i)  $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$$(2x^2 - 3x - 4 = 0)$$

$$x = \frac{3 \pm \sqrt{9 - 4(2)(-4)}}{2(2)}$$

$$x = \frac{3 \pm \sqrt{41}}{4}$$

$$x = -0.85, 2.35 \quad \text{to 2 decimal places}$$

ii)  $\Delta = b^2 - 4ac$

$$(2x^2 - 3x + 4 = 0)$$

$$\Delta = (-3)^2 - 4(2)(4)$$

$$\Delta = -23$$

since  $\Delta < 0$ ,  $2x^2 - 3x + 4 = 0$  has no real roots.

d) i)  $\int x^6 dx = \frac{x^7}{7} + C$

ii)  $\int (x-1)(x+2) dx = \int (x^2 + x - 2) dx$   
 $= \frac{x^3}{3} + \frac{x^2}{2} - 2x + C$

$$\begin{aligned}
 \text{iii)} \quad \int \frac{\sqrt{t}+1}{\sqrt{t}} dt &= \int (1+t^{-\frac{1}{2}}) dt \\
 &= t + \frac{t^{\frac{1}{2}}}{(\frac{1}{2})} + C \\
 &= t + 2\sqrt{t} + C
 \end{aligned}$$

e) let the three terms be  $a-d, a, a+d$

$$(a-d) + (a) + (a+d) = 21$$

$$3a = 21$$

$$a = 7$$

$$(a-d)(a)(a+d) = 315$$

$$a(a^2 - d^2) = 315$$

$$\text{sub in } a=7$$

$$7(49 - d^2) = 315$$

$$49 - d^2 = 45$$

$$-d^2 = -4$$

$$d^2 = 4$$

$$d = \pm 2$$

$\therefore$  the three terms are 5, 7 & 9.

Q2.

(a) (i)  $ab^c = d$

$$b^c = \frac{d}{a}$$

$$\left| b = \left( \frac{d}{a} \right)^{\frac{1}{c}} \right| \text{ OR } \sqrt[c]{d/a}.$$

(ii)  $b^c = \frac{d}{a}$

$$\left| c = \log_b \frac{d}{a} \right|$$

(iii)  $b = \left( \frac{61.94}{75.12} \right)^{\frac{1}{1.142}}$

$$\left| \approx 0.8446 \right| \text{ (4.S.F.)}$$

(b) (i) Let the roots be  $\alpha, \alpha-3$ .

$$\text{now } \alpha + \alpha - 3 = -2 \quad (S_1 = -\frac{b}{a})$$

$$2\alpha = 1$$

$$\alpha = \frac{1}{2}$$

$$\text{now } S_2 = \frac{c}{a} = -a$$

$$\therefore a = -\left( \frac{1}{2} \times -\frac{5}{2} \right)$$

$$\left| a = \frac{5}{4} \right|$$

(ii) If  $f(x) < 0$  the quadratic would be negative definite.

$$\therefore \Delta < 0$$

$$\text{ie } (-2)^2 - 4 \times 1 \times a < 0$$

$$4 + 4a < 0$$

$$4a < -4$$

$$\left| a < -1 \right|$$

$$(c) \quad f(x) = 6x^2 - 6x.$$

$$\text{let } f'(x) = 0$$

$$6x(x-1) = 0$$

$$x = 0, 1.$$

$$\therefore \boxed{\text{Points are } (0, 1) \text{ and } (1, 0)}$$

NOTE If  $x = 0$

$$f(0) = 0 - 0 + 1 = 1$$

If  $x = 1$

$$f(1) = 2 - 3 + 1 = 0.$$

$$(d) \quad \int_{-1}^2 (3x^2 - 2x) dx = \left[ x^3 - x^2 \right]_{-1}^2$$

$$= (8 - 4) - (1 - 1)$$

$$= 4 - 0$$

$$= 4.$$

(NB Answer is not 6)

$$(ex) \quad f(1) = 5 - 7 + 3 + 2 = 3.$$

$$\therefore (1, 3) \text{ lies on } f(x) = 5x^3 - 7x^2 + 3x + 2.$$

$$(ii) \quad f(x) = 5x^3 - 7x^2 + 3x + 2.$$

$$f'(x) = 15x^2 - 14x + 3$$

$$\therefore f'(1) = 15 - 14 + 3$$

$$= 4$$

$$\therefore \boxed{\text{gradient at } (1, 3) \text{ is } 4}$$

$$(iii) \quad y - 3 = 4(x - 1)$$

$$y - 3 = 4x - 4$$

$$\boxed{y = 4x - 1}$$

in "y = mx + b" form.

$$(iii) \quad y - 3 = -\frac{1}{4}(x - 1)$$

$$4y - 12 = x - 1$$

$$\boxed{x - 4y + 13 = 0}$$

in "general form"

**Question 3 (15 Marks)**

- (a) Find the values of  $y$  which satisfy the equation:

2

$$(8^y)^y \times \frac{1}{32^y} = 4$$

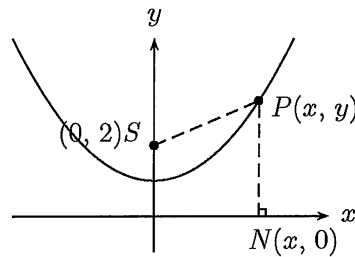
**Solution:**

$$\begin{aligned} 2^{3y^2} \times 2^{-5y} &= 2^2, \\ 3y^2 - 5y - 2 &= 0, \\ (3y + 1)(y - 2) &= 0, \\ \therefore y &= 2, -\frac{1}{3}. \end{aligned}$$

- (b) A point  $P$  is equidistant from the  $x$ -axis and the point  $F(0, 2)$ . Find the locus of the point  $P$ .

3

**Solution:**



$$\begin{aligned} PN^2 &= PS^2, \\ y^2 &= x^2 + (y - 2)^2, \\ y^2 &= x^2 + y^2 - 4y + 4, \\ \therefore x^2 &= 4(y - 1). \end{aligned}$$

- (c) Using first principles, find the derivative of the function  $f(x) = x^2 + x$  (all working must be shown).

2

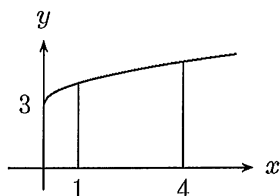
**Solution:**

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \left\{ \frac{(x+h)^2 + (x+h) - x^2 - x}{h} \right\}, \\ &= \lim_{h \rightarrow 0} \left\{ \frac{x^2 + 2xh + h^2 + x + h^2 - x}{h} \right\}, \\ &= \lim_{h \rightarrow 0} \left\{ \frac{2xh + h^2 + h}{h} \right\}, \\ &= \lim_{h \rightarrow 0} \{2x + h + 1\}, \\ &= 2x + 1. \end{aligned}$$

- (d) Find the area bounded by  $y = \sqrt{x} + 3$  and the  $x$ -axis for  $1 \leq x \leq 4$ .

2

**Solution:**



$$\begin{aligned}
 \text{Area} &= \int_1^4 (x^{\frac{1}{2}} + 3) dx, \\
 &= \left[ \frac{2x^{\frac{3}{2}}}{3} + 3x \right]_1^4, \\
 &= \frac{16}{3} + 12 - \left( \frac{2}{3} + 3 \right), \\
 &= \frac{41}{3} \text{ or } 13\frac{2}{3}.
 \end{aligned}$$

- (e) The equation  $3x^2 - 6x + 8 = 0$  has roots  $\alpha$  and  $\beta$ . Find an equation which has roots  $\frac{1}{\alpha}$  and  $\frac{1}{\beta}$ .

2

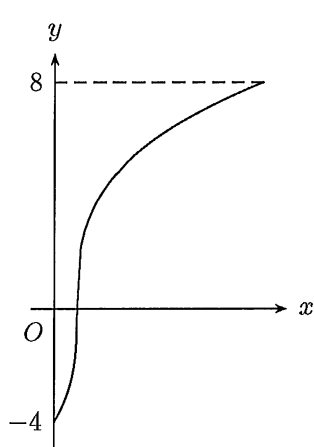
$$\begin{aligned}
 \text{Solution: } \alpha + \beta &= \frac{2}{3}, & \frac{1}{\alpha} + \frac{1}{\beta} &= \frac{\alpha + \beta}{\alpha\beta}, \\
 \alpha\beta &= \frac{8}{3}. & &= \frac{\frac{2}{3}}{\frac{8}{3}}, \\
 & & \frac{1}{\alpha\beta} &= \frac{3}{8}.
 \end{aligned}$$

$$\begin{aligned}
 \therefore \text{New equation: } x^2 - \frac{3x}{4} + \frac{3}{8} &= 0, \\
 \text{i.e. } 8x^2 - 6x + 3 &= 0.
 \end{aligned}$$

- (f) A wall vase has one plane face, and its volume is equivalent to that generated when the area enclosed by  $x = \frac{y^3}{64} + 1$ , the  $y$ -axis, and  $y = 8$ , is rotated through two right angles about the  $y$ -axis, the units being centimetres. Calculate its volume.

4

**Solution:** When  $x = 0$ ,  $y = -4$ .



$$\begin{aligned}
 V &= \frac{1}{2}\pi \int_{-4}^8 x^2 dy, \\
 &= \frac{\pi}{2} \int_{-4}^8 \left( \frac{y^6}{4096} + \frac{y^3}{32} + 1 \right) dy, \\
 &= \frac{\pi}{2} \left[ \frac{y^7}{7 \times 4096} + \frac{y^4}{4 \times 32} + y \right]_{-4}^8, \\
 &= \frac{\pi}{2} \left\{ \frac{512}{7} + 32 + 8 - \left( -\frac{4}{7} + 2 - 4 \right) \right\}, \\
 &= \frac{405\pi}{7}, \\
 &\approx 181.76 \text{ cm}^3 \text{ (2 dec. pl.)}
 \end{aligned}$$

# QUESTION 4

a (i)

$$f(x) = \sqrt{1+x^2}$$

|        |            |             |             |             |
|--------|------------|-------------|-------------|-------------|
| $x$    | 2          | 3           | 4           | 5           |
| $f(x)$ | $\sqrt{5}$ | $\sqrt{10}$ | $\sqrt{17}$ | $\sqrt{26}$ |
|        | 2.24       | 3.16        | 4.12        | 5.10        |

(ii)

$$A = \frac{h}{2} \{ f(2) + f(5) + 2[f(3) + f(4)] \}$$

$$h = \frac{b-a}{n} = \frac{5-2}{3} = 1$$

$$A = \frac{1}{2} [2.24 + 5.10 + 2(3.16 + 4.12)]$$

$$= 10.950 \quad (3 \text{ d.p.})$$

b (i)

$$P(a, 3a-9) \quad Q(1, 4)$$

$$PQ^2 = \sqrt{(a-1)^2 + (3a-9-4)^2}$$

$$PQ^2 = a^2 - 2a + 1 + 9a^2 - 78a + 169$$

$$PQ^2 = 10a^2 - 80a + 170$$

(ii)

$$(PQ^2)' = 20a - 80$$

$$\text{turn pt. } 20a = 80, a = 4$$

$$(PQ^2)'' = 20$$

$$(PQ^2)'' > 0$$

$$\therefore \text{min value of } PQ = 4$$

(iii)

min value when line  $\perp$  to PQ

$$\therefore N \text{ is } (4, 3)$$

(iv)

$$y - 3 = -\frac{1}{3}(x - 4)$$

$$3y - 9 = -x + 4$$

$$x + 3y - 13 = 0$$

# SECTION E

# QUESTION 5

a) (i)  $\log_5 x + \log_2 8 = 0$

$$\log_5 x = -3$$

$$x = 5^{-3} = \frac{1}{125}$$

(ii)  $\log_3 x + \frac{\log_3 27}{\log_3 x} = 4$

$$(\log_3 x)^2 - 4 \log_3 x + 3 = 0$$

$$\log_3 x = 3 \quad \text{or} \quad \log_3 x = 1$$

$$x = 27 \quad \text{or} \quad x = 3$$

b)  $r_1 + r_2 = k$  — (\*)

$$S = \pi r_1^2 + \pi r_2^2$$

$$= \pi (r_1^2 + r_2^2)$$

$$= \pi [r_1^2 + (k - r_1)^2]$$

$$S = \pi [2r_1^2 - 2kr_1 + k^2]$$

$$\frac{dS}{dr_1} = \pi [4r_1 - 2k] = 0$$

$$\Rightarrow r_1 = k/2$$

$$\therefore r_2 = k/2 \quad \text{from (*)}$$

$$\frac{d^2 S}{dr_1^2} = 4\pi > 0 \quad \text{MIN.}$$

(c)  $V_n$  = value at end of  $n$  years (after  $n^{\text{th}}$  prize).

(i)  $V_1 = 2000(1.025)^2 - 150$

$$V_1 = \$1951.25$$

(ii)  $V_1 = 2000(1.025)^2 - 150$

$$V_2 = [2000(1.025)^2 - 150](1.025)^2 - 150$$

$$V_2 = 2000(1.025)^4 - 150[1 + 1.025^2]$$

$$V_3 = \{2000(1.025)^4 - 150[1 + 1.025^2]\}(1.025^2 - 150)$$

$$V_3 = 2000(1.025)^6 - 150[1 + 1.025^2 + 1.025^4]$$

$$V_3 = \$1846.77$$

(iii)

$$V_n = 2000(1.025)^{2n} - 150[1 + 1.025^2 + 1.025^4 + \dots]$$

$$\text{let } V_n = 0$$

$$\frac{2000(1.025)^{2n}}{150} = \frac{1(1.025^2)^n - 1}{0.050625}$$

$$0.675(1.025)^{2n} = 1.025^{2n} - 1$$

$$1 = [1.025^{2n}](0.325)$$

$$n = \frac{-1}{2} \cdot \frac{\log(0.325)}{\log(1.025)} \approx 22$$

22 prizes

(d))

(i)  $a, b, c$  A.S.

$a, c, b$  G.S.

$$\Rightarrow b - a = c - b$$

$$b = \frac{a+c}{2}$$

$$\boxed{r = \frac{c}{a} = \frac{b}{c}}$$

$$c^2 = ab$$

$$\Rightarrow c^2 = a \left( \frac{a+c}{2} \right)$$

$$2c^2 = a^2 + ac$$

$$2c^2 - ac - a^2 = 0$$

$$\left( \div a^2 \right)$$

$$2\left(\frac{c}{a}\right)^2 - \left(\frac{c}{a}\right) - 1 = 0$$

$$\Rightarrow \frac{c}{a} = \frac{1 \pm \sqrt{1 - 4(2)(-1)}}{4}$$

Common ratio  $\frac{c}{a} = 1$  or  $-\frac{1}{2}$

However only  $-\frac{1}{2}$  applies  $\therefore \boxed{r = -\frac{1}{2}}$

$$(ii) S = \frac{a}{1-r}$$

$$= \frac{a}{1 - (-\frac{1}{2})}$$

$$\boxed{S = \frac{2}{3}a}$$